

Hot Finished Rectangular Hollow Sections

BS EN 10210

Designation		Weight	Cross Sectional Area	Second Moment of Area	Second Moment of Area	Radius of Gyratation	Radius of Gyratation	Elastic Modulus	Elastic Modulus	Plastic Modulus	Plastic Modulus	Torsional inertia contant	Torsional modulus contant	Superficial Area per metre length	Nominal Length per tonne	Ratio for Local Buckling	Ratio for Local Buckling
Size	Thickness																
B × H	T	M	A	I _{yy}	I _{zz}	i _{yy}	i _{zz}	W _{el_{yy}}	W _{el_{zz}}	W _{pl_{yy}}	W _{pl_{zz}}	L _t	C _t	As	L	Flange	Web
mm	mm	Kg/m	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³	cm ³	cm ³	cm ⁴	cm ³	m ² /m	m	B/T	H/T
19 × 38	1.6	1.330	1.69	3.03	1.00	1.34	0.77	1.60	1.05	2.02	1.23	2.45	1.79	0.11	751.94	7.88	19.75
20 × 40	2.0	1.725	2.20	4.26	1.39	1.39	0.80	2.13	1.39	2.73	1.65	3.45	2.36	0.11	579.81	6.00	16.00
	2.5	2.106	2.68	5.02	1.61	1.37	0.77	2.51	1.61	3.27	1.96	4.07	2.73	0.11	474.81	4.00	12.00
	3.0	2.468	3.14	5.66	1.78	1.34	0.75	2.83	1.78	3.75	2.23	4.59	3.03	0.11	405.25	2.67	9.33
	1.6	1.481	1.89	3.67	1.90	1.40	1.00	1.93	1.52	2.37	1.77	3.98	2.46	0.12	675.40	11.63	19.75
25 × 50	1.6	1.782	2.27	7.24	2.42	1.79	1.03	2.90	1.94	3.62	2.22	5.84	3.30	0.15	561.15	11.63	27.25
	2.0	2.196	2.80	8.73	2.89	1.77	1.02	3.49	2.31	4.41	2.68	7.05	3.93	0.14	455.43	8.50	21.00
	2.3	2.498	3.18	9.76	3.20	1.75	1.00	3.90	2.56	4.97	3.01	7.89	4.35	0.14	400.39	6.87	17.74
	2.5	2.695	3.43	10.41	3.39	1.74	0.99	4.16	2.71	5.33	3.22	8.42	4.61	0.14	371.08	6.00	16.00
	3.0	3.174	4.04	11.90	3.83	1.72	0.97	4.76	3.06	6.18	3.71	9.64	5.20	0.14	315.05	4.33	12.67
25 × 75	1.6	2.410	3.07	20.25	3.52	2.57	1.07	5.40	2.82	6.96	3.15	9.91	5.05	0.20	414.93	11.63	42.88
	1.9	2.839	3.62	23.56	4.05	2.55	1.06	6.28	3.24	8.14	3.67	11.48	5.81	0.20	352.21	9.16	35.47
	2.0	2.981	3.80	24.63	4.21	2.55	1.05	6.57	3.37	8.53	3.83	11.99	6.04	0.19	335.49	8.50	33.50
	2.3	3.400	4.33	27.74	4.69	2.53	1.04	7.40	3.75	9.66	4.32	13.44	6.73	0.19	294.09	6.87	28.61
	2.5	3.676	4.68	29.74	4.98	2.52	1.03	7.93	3.99	10.40	4.63	14.36	7.15	0.19	272.03	6.00	26.00
	3.0	4.352	5.54	34.44	5.65	2.49	1.01	9.19	4.52	12.17	5.36	16.49	8.12	0.19	229.80	4.33	21.00
	4.0	5.643	7.19	42.70	6.71	2.44	0.97	11.39	5.37	15.41	6.64	20.07	9.67	0.19	177.22	2.25	14.75
	4.5	6.259	7.97	46.27	7.12	2.41	0.95	12.34	5.70	16.88	7.19	21.53	10.27	0.19	159.78	1.56	12.67
30 × 50	2.0	2.353	3.00	9.88	4.41	1.82	1.21	3.95	2.94	4.89	3.41	9.74	4.84	0.15	425.04	11.00	21.00
	2.5	2.891	3.68	11.82	5.22	1.79	1.19	4.73	3.48	5.92	4.11	11.72	5.73	0.15	345.89	8.00	16.00
	3.0	3.410	4.34	13.56	5.94	1.77	1.17	5.43	3.96	6.88	4.76	13.52	6.51	0.15	293.29	6.00	12.67
	4.0	4.387	5.59	16.49	7.08	1.72	1.13	6.60	4.72	8.59	5.88	16.59	7.77	0.15	227.96	3.50	8.50
	5.0	5.284	6.73	18.71	7.89	1.67	1.08	7.49	5.26	10.03	6.80	18.97	8.67	0.15	189.24	2.00	6.00
30 × 60	1.6	2.159	2.75	12.82	4.32	2.16	1.25	4.27	2.88	5.30	3.26	10.33	4.90	0.18	463.21	14.75	33.50
	1.9	2.541	3.24	14.88	4.98	2.14	1.24	4.96	3.32	6.19	3.79	12.00	5.65	0.18	393.56	11.79	27.58
	2.3	3.039	3.87	17.47	5.79	2.12	1.22	5.82	3.86	7.34	4.47	14.11	6.57	0.17	329.03	9.04	22.09
	3.0	3.881	4.94	21.58	7.04	2.09	1.19	7.19	4.69	9.20	5.57	17.46	7.97	0.17	257.69	6.00	16.00
30 × 80	1.6	2.661	3.39	26.38	5.61	2.79	1.29	6.60	3.74	8.37	4.17	15.08	6.63	0.22	375.76	14.75	46.00
	1.9	3.138	4.00	30.76	6.49	2.77	1.27	7.69	4.32	9.81	4.86	17.54	7.65	0.22	318.72	11.79	38.11
	2.3	3.761	4.79	36.32	7.56	2.75	1.26	9.08	5.04	11.67	5.75	20.65	8.92	0.21	265.86	9.04	30.78
	3.0	4.823	6.14	45.33	9.23	2.72	1.23	11.33	6.16	14.75	7.19	25.62	10.90	0.21	207.36	6.00	22.67
30 × 110	2.0	4.237	5.40	73.36	9.12	3.69	1.30	13.34	6.08	17.48	6.77	27.30	11.12	0.27	236.03	11.00	51.00
	2.5	5.246	6.68	89.48	10.91	3.66	1.28	16.27	7.27	21.47	8.24	32.98	13.31	0.27	190.62	8.00	40.00
	3.0	6.236	7.94	104.74	12.53	3.63	1.26	19.04	8.35	25.31	9.62	38.21	15.28	0.27	160.37	6.00	32.67
	4.0	8.155	10.39	132.74	15.26	3.57	1.21	24.13	10.17	32.56	12.12	47.39	18.65	0.27	122.63	3.50	23.50
35 × 65	1.6	2.410	3.07	17.25	6.57	2.37	1.46	5.31	3.76	6.51	4.23	15.07	6.32	0.20	414.93	17.88	36.63
	1.9	2.839	3.62	20.08	7.61	2.36	1.45	6.18	4.35	7.63	4.94	17.57	7.31	0.20	352.21	14.42	30.21
	2.0	2.981	3.80	20.99	7.94	2.35	1.45	6.46	4.54	7.99	5.17	18.38	7.63	0.19	335.49	13.50	28.50
	2.3	3.400	4.33	23.65	8.89	2.34	1.43	7.28	5.08	9.05	5.85	20.74	8.54	0.19	294.09	11.22	24.26
	2.5	3.676	4.68	25.35	9.50	2.33	1.42	7.80	5.43	9.74	6.28	22.26	9.12	0.19	272.03	10.00	22.00
	3.0	4.352	5.54	29.38	10.91	2.30	1.40	9.04	6.23	11.41	7.32	25.86	10.46	0.19	229.80	7.67	17.67
	4.0	5.643	7.19	36.47	13.28	2.25	1.36	11.22	7.59	14.45	9.19	32.25	12.74	0.19	177.22	4.75	12.25
38 × 65	1.6	2.485	3.17	18.22	7.91	2.40	1.58	5.60	4.16	6.82	4.70	17.35	6.91	0.20	402.35	19.75	36.63
	1.9	2.929	3.73	21.21	9.17	2.38	1.57	6.53	4.83	7.99	5.49	20.25	8.01	0.20	341.45	16.00	30.21
	2.0	3.075	3.92	22.18	9.58	2.38	1.56	6.82	5.04	8.37	5.75	21.19	8.36	0.20	325.21	15.00	28.50
	2.3	3.509	4.47	25.01	10.75	2.37	1.55	7.69	5.66	9.49	6.51	23.95	9.37	0.20	285.01	12.52	24.26
	2.5	3.794	4.83	26.82	11.49	2.36	1.54	8.25	6.05	10.21	7.00	25.73	10.01	0.20	263.58	11.20	22.00
	3.0	4.493	5.72	31.11	13.23	2.33	1.52	9.57	6.96	11.96	8.17	29.96	11.52	0.20	222.57	8.67	17.67
	3.2	4.767	6.07	32.74	13.88	2.32	1.51	10.07	7.30	12.64	8.61	31.57	12.08	0.20	209.78	7.88	16.31
	4.0	5.831	7.43	38.71	16.20	2.28	1.48	11.91	8.53	15.19	10.29	37.54	14.10	0.20	171.49	5.50	12.25
38 × 75	1.6	2.737	3.49	25.85	8.97	2.72	1.60	6.89	4.72	8.48	5.28	21.11	8.03	0.22	365.42	19.75	42.88
	1.9	3.227	4.11	30.16	10.41	2.71	1.59	8.04	5.48	9.95	6.18	24.65	9.31	0.22	309.88	16.00	35.47
	2.0	3.389	4.32	31.56	10.87	2.70	1.59	8.42	5.72	10.42	6.47	25.81	9.72	0.22	295.08	15.00	33.50
	2.3	3.870	4.93	35.65	12.22	2.69	1.57	9.51	6.43	11.84	7.33	29.18	10.91	0.22	258.41	12.52	28.61
	2.5	4.186	5.33	38.28	13.07	2.68	1.57	10.21	6.88	12							

Hot Finished Rectangular Hollow Sections

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Designation		Weight	Cross Sectional Area	Second Moment of Area	Second Moment of Area	Radius of Gyration	Radius of Gyration	Elastic Modulus	Elastic Modulus	Plastic Modulus	Plastic Modulus	Torsional Inertia contant	Torsional modulus contant	Superficial Area per metre length	Nominal Length per tonne	Ratio for	Ratio for
Size	Thickness															Local Buckling	Local Buckling
$B \times H$	T	M	A	I_{yy}	I_{zz}	i_{yy}	i_{zz}	W_{elzy}	W_{elxz}	W_{plzy}	W_{plxz}	L_t	C_t	A_s	L	Flange	Web
mm	mm	Kg/m	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³	cm ³	cm ³	cm ⁴	cm ³	m ² /m	m	B/T	H/T
50 × 90	2.0	4.237	5.40	59.03	23.71	3.31	2.10	13.12	9.48	16.00	10.65	53.25	15.89	0.27	236.03	21.00	41.00
	2.5	5.246	6.68	72.04	28.76	3.28	2.07	16.01	11.50	19.66	13.04	65.15	19.25	0.27	190.62	16.00	32.00
	3.0	6.236	7.94	84.38	33.47	3.26	2.05	18.75	13.39	23.19	15.34	76.49	22.38	0.27	160.37	12.67	26.00
	4.0	8.155	10.39	107.09	41.95	3.21	2.01	23.80	16.78	29.85	19.63	97.52	28.02	0.27	122.63	8.50	18.50
	5.0	9.994	12.73	127.26	49.21	3.16	1.97	28.28	19.69	35.99	23.53	116.38	32.86	0.27	100.06	6.00	14.00
50 × 100	1.6	3.666	4.67	62.22	21.30	3.65	2.14	12.44	8.52	15.23	9.43	50.00	14.53	0.30	272.77	27.25	58.50
	1.9	4.331	5.52	72.92	24.86	3.64	2.12	14.58	9.94	17.91	11.07	58.64	16.95	0.30	230.91	22.32	48.63
	2.0	4.551	5.80	76.41	26.02	3.63	2.12	15.28	10.41	18.80	11.61	61.46	17.73	0.29	219.75	21.00	46.00
	2.3	5.206	6.63	86.71	29.40	3.62	2.11	17.34	11.76	21.41	13.19	69.80	20.03	0.29	192.09	17.74	39.48
	2.5	5.639	7.18	93.42	31.58	3.61	2.10	18.68	12.63	23.13	14.23	75.23	21.50	0.29	177.35	16.00	36.00
	3.0	6.707	8.54	109.60	36.79	3.58	2.08	21.92	14.72	27.31	16.75	88.37	25.03	0.29	149.11	12.67	29.33
	4.0	8.783	11.19	139.60	46.19	3.53	2.03	27.92	18.48	35.24	21.47	112.78	31.40	0.29	113.86	8.50	21.00
	4.5	9.791	12.47	153.44	50.40	3.51	2.01	30.69	20.16	39.00	23.68	124.06	34.26	0.29	102.13	7.11	18.22
	5.0	10.779	13.73	166.52	54.30	3.48	1.99	33.30	21.72	42.61	25.78	134.74	36.91	0.29	92.77	6.00	16.00
	6.0	12.696	16.17	190.46	61.20	3.43	1.95	38.09	24.48	49.41	29.68	154.30	41.62	0.28	78.76	4.33	12.67
	6.3	13.256	16.89	197.08	63.05	3.42	1.93	39.42	25.22	51.35	30.77	159.71	42.89	0.28	75.44	4.94	12.87
50 × 125	2.3	6.109	7.78	151.21	35.94	4.41	2.15	24.19	14.38	30.42	15.94	93.74	25.26	0.34	163.70	17.74	50.35
	2.5	6.620	8.43	163.11	38.64	4.40	2.14	26.10	15.45	32.89	17.20	101.06	27.15	0.34	151.06	16.00	46.00
	3.0	7.884	10.04	192.01	45.08	4.37	2.12	30.72	18.03	38.93	20.28	118.79	31.67	0.34	126.84	12.67	37.67
	4.0	10.353	13.19	246.23	56.79	4.32	2.08	39.40	22.72	50.48	26.07	151.85	39.88	0.34	96.59	8.50	27.25
	4.5	11.557	14.72	271.59	62.08	4.29	2.05	43.45	24.83	55.99	28.80	167.18	43.59	0.34	86.53	7.11	23.78
	5.0	12.742	16.23	295.79	67.01	4.27	2.03	47.33	26.80	61.34	31.40	181.74	47.06	0.34	78.48	6.00	21.00
	6.0	15.051	19.17	340.83	75.81	4.22	1.99	54.53	30.32	71.51	36.28	208.53	53.28	0.33	66.44	4.33	16.83
	6.3	15.729	20.04	353.48	78.19	4.20	1.98	56.56	31.28	74.42	37.65	215.97	54.97	0.33	63.58	4.94	16.84
50 × 150	2.3	7.011	8.93	240.02	42.49	5.18	2.18	32.00	17.00	40.87	18.68	118.26	30.50	0.39	142.63	17.74	61.22
	3.0	9.062	11.54	305.80	53.38	5.15	2.15	40.77	21.35	52.42	23.80	149.94	38.30	0.39	110.36	12.67	46.00
	4.0	11.923	15.19	394.08	67.40	5.09	2.11	52.54	26.96	68.21	30.67	191.81	48.36	0.39	83.87	8.50	33.50
	4.5	13.324	16.97	435.75	73.76	5.07	2.08	58.10	29.51	75.80	33.91	211.27	52.93	0.39	75.05	7.11	29.33
	5.0	14.704	18.73	475.80	79.71	5.04	2.06	63.44	31.89	83.19	37.03	229.77	57.22	0.39	68.01	6.00	26.00
	6.0	17.406	22.17	551.12	90.42	4.99	2.02	73.48	36.17	97.35	42.88	263.92	64.95	0.38	57.45	4.33	21.00
	6.3	18.201	23.19	572.49	93.33	4.97	2.01	76.33	37.33	101.44	44.54	273.43	67.07	0.38	54.94	4.94	20.81
60 × 80	2.0	4.237	5.40	50.43	32.37	3.06	2.45	12.61	10.79	14.96	12.28	61.09	17.08	0.27	236.03	26.00	36.00
	2.5	5.246	6.68	61.52	39.38	3.03	2.43	15.38	13.13	18.38	15.07	74.90	20.74	0.27	190.62	20.00	28.00
	3.0	6.236	7.94	72.02	45.97	3.01	2.41	18.01	15.32	21.68	17.75	88.13	24.17	0.27	160.37	16.00	22.67
	4.0	8.155	10.39	91.33	57.96	2.97	2.36	22.83	19.32	27.90	22.79	112.87	30.38	0.27	122.63	11.00	16.00
	5.0	9.994	12.73	108.45	68.43	2.92	2.32	27.11	22.81	33.63	27.39	135.35	35.78	0.27	100.06	8.00	12.00
60 × 100	2.5	6.031	7.68	105.30	47.65	3.70	2.49	21.06	15.88	25.56	17.95	103.02	26.25	0.31	165.81	20.00	36.00
	3.0	7.178	9.14	123.72	55.73	3.68	2.47	24.74	18.58	30.22	21.17	121.39	30.66	0.31	139.32	16.00	29.33
	4.0	9.411	11.99	158.04	70.52	3.63	2.43	31.61	23.51	39.08	27.27	155.92	38.74	0.31	106.26	11.00	21.00
	5.0	11.564	14.73	189.10	83.59	3.58	2.38	37.82	27.86	47.36	32.89	187.55	45.86	0.31	86.47	8.00	16.00
	6.0	13.638	17.37	217.01	95.02	3.53	2.34	43.40	31.67	55.05	38.07	216.32	52.10	0.30	73.32	6.00	12.67
	6.3	14.245	18.15	224.78	98.15	3.52	2.33	44.96	32.72	57.25	39.53	224.39	53.81	0.30	70.20	6.52	12.87
60 × 120	2.5	6.816	8.68	164.45	55.92	4.35	2.54	27.41	18.64	33.75	20.82	132.30	31.76	0.35	146.71	20.00	44.00
	3.0	8.120	10.34	193.71	65.48	4.33	2.52	32.28	21.83	39.96	24.59	156.00	37.16	0.35	123.16	16.00	36.00
	4.0	10.667	13.59	248.73	83.09	4.28	2.47	41.46	27.70	51.87	31.75	200.68	47.10	0.35	93.75	11.00	26.00
	4.5	11.911	15.17	274.53	91.16	4.25	2.45	45.76	30.39	57.57	35.13	221.68	51.66	0.35	83.96	9.33	22.67
	5.0	13.134	16.73	299.21	98.76	4.23	2.43	49.87	32.92	63.09	38.39	241.80	55.95	0.35	76.14	8.00	20.00
	6.0	15.522	19.77	345.29	112.59	4.18	2.39	57.55	37.53	73.63	44.55	279.40	63.78	0.34	64.42	6.00	16.00
	6.3	16.223	20.67	358.27	116.40	4.16	2.37	59.71	38.80	76.66	46.30	290.01	65.94	0.34	61.64	6.52	16.05
75 × 100	2.3	6.109	7.78	114.16	73.34	3.83	3.07	22.83	19.56	27.03	22.20	138.05	30.99	0.34	163.70	28.61	39.48
	3.0	7.884	10.04	144.90	92.80	3.80	3.04	28.98	24.75	34.59	28.37	176.22	39.11	0.34	126.84	21.00	29.33
	3.2	8.384	10.68	153.35	98.12	3.79	3.03	30.67	26.17	36.69	30.08	186.80	41.33	0.34	119.27	19.44	27.25
	4.0	10.353	13.19	185.71	118.39	3.75	3.00	37.14	31.57	44.84	36.71	227.68	49.76	0.34	96.59		

Hot Finished Rectangular Hollow Sections

BS EN 10210

Designation		Weight	Cross Sectional Area	Second Moment of Area	Second Moment of Area	Radius of Gyration	Radius of Gyration	Elastic Modulus	Elastic Modulus	Plastic Modulus	Plastic Modulus	Torsional Inertia constant	Torsional modulus constant	Superficial Area per metre length	Nominal Length per tonne	Ratio for Local Buckling	Ratio for Local Buckling
Size	Thickness																
$B \times H$	T	M	A	I_{yy}	I_{zz}	i_{yy}	i_{zz}	$W_{el yy}$	$W_{el zz}$	$W_{pl yy}$	$W_{pl zz}$	L_t	C_t	A_s	L	Flange	Web
mm	mm	Kg/m	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³	cm ³	cm ³	cm ⁴	cm ³	m ² /m	m	B/T	H/T
80 × 140	4.0	13.179	16.79	440.60	183.83	5.12	3.31	62.94	45.96	77.14	52.22	410.65	76.53	0.43	75.88	16.00	31.00
	5.0	16.274	20.73	533.96	221.11	5.08	3.27	76.28	55.28	94.32	63.63	499.36	91.94	0.43	61.45	12.00	24.00
	6.0	19.290	24.57	620.92	255.18	5.03	3.22	88.70	63.80	110.68	74.40	582.61	106.01	0.42	51.84	9.33	19.33
	6.3	20.180	25.71	645.79	264.80	5.01	3.21	92.26	66.20	115.43	77.51	606.52	109.98	0.42	49.55	9.70	19.22
	8.0	25.083	31.95	776.32	314.20	4.93	3.14	110.90	78.55	140.97	94.07	732.93	130.43	0.42	39.87	7.00	14.50
80 × 160	3.0	10.946	13.94	472.09	161.00	5.82	3.40	59.01	40.25	72.47	44.78	379.61	68.61	0.47	91.36	22.67	49.33
	3.2	11.650	14.84	500.79	170.49	5.81	3.39	62.60	42.62	77.00	47.54	402.81	72.63	0.47	85.84	21.00	46.00
	4.0	14.435	18.39	612.21	206.96	5.77	3.35	76.53	51.74	94.73	58.30	493.03	88.08	0.47	69.28	16.00	36.00
	4.5	16.150	20.57	679.13	228.56	5.75	3.33	84.89	57.14	105.51	64.80	547.32	97.23	0.47	61.92	13.78	31.56
	5.0	17.844	22.73	744.00	249.28	5.72	3.31	93.00	62.32	116.05	71.13	600.01	106.01	0.47	56.04	12.00	28.00
	6.0	21.174	26.97	867.66	288.11	5.67	3.27	108.46	72.03	136.45	83.28	700.62	122.45	0.46	47.23	9.33	22.67
	6.3	22.158	28.23	903.20	299.10	5.66	3.26	112.90	74.78	142.40	86.80	729.58	127.11	0.46	45.13	9.70	22.40
	8.0	27.595	35.15	1091.28	355.84	5.57	3.18	136.41	88.96	174.52	105.59	883.05	151.19	0.46	36.24	7.00	17.00
	9.0	30.686	39.09	1191.60	385.01	5.52	3.14	148.95	96.25	192.21	115.77	965.00	163.62	0.46	32.59	5.89	14.78
100 × 150	3.0	11.417	14.54	467.89	250.78	5.67	4.15	62.39	50.16	74.47	56.41	506.22	81.43	0.49	87.59	29.33	46.00
	4.0	15.063	19.19	607.29	324.03	5.63	4.11	80.97	64.81	97.41	73.64	660.13	104.99	0.49	66.39	21.00	33.50
	4.5	16.856	21.47	673.99	358.79	5.60	4.09	89.87	71.76	108.54	81.97	734.34	116.16	0.49	59.33	18.22	29.33
	5.0	18.629	23.73	738.71	392.35	5.58	4.07	98.50	78.47	119.44	90.11	806.73	126.91	0.49	53.68	16.00	26.00
	6.0	22.116	28.17	862.34	455.90	5.53	4.02	114.98	91.18	140.55	105.81	946.07	147.26	0.48	45.22	12.67	21.00
	6.3	23.147	29.49	897.93	474.06	5.52	4.01	119.72	94.81	146.70	110.38	986.47	153.06	0.48	43.20	12.87	20.81
	8.0	28.851	36.75	1086.89	569.30	5.44	3.94	144.92	113.86	180.07	134.98	1203.23	183.49	0.48	34.66	9.50	15.75
	9.0	32.099	40.89	1188.18	619.45	5.39	3.89	158.42	123.89	198.50	148.47	1321.15	199.51	0.48	31.15	8.11	13.67
	12.5	48.728	62.07	2384.91	907.60	6.20	3.82	264.99	181.52	343.65	222.86	2191.20	303.07	0.53	20.52	5.50	11.90
100 × 180	4.0	16.947	21.59	944.51	379.35	6.61	4.19	104.95	75.87	128.00	85.16	852.07	127.11	0.55	59.01	21.00	41.00
	5.0	20.984	26.73	1152.67	460.10	6.57	4.15	128.07	92.02	157.29	104.36	1042.42	153.99	0.55	47.65	16.00	32.00
	6.0	24.942	31.77	1350.07	535.53	6.52	4.11	150.01	107.11	185.51	122.73	1223.86	179.06	0.54	40.09	12.67	26.00
	6.3	26.114	33.27	1407.22	557.15	6.50	4.09	156.36	111.43	193.77	128.09	1276.57	186.25	0.54	38.29	12.87	25.57
	8.0	32.619	41.55	1713.39	671.13	6.42	4.02	190.38	134.23	238.80	157.06	1560.32	224.13	0.54	30.66	9.50	19.50
	10.0	39.978	50.93	2036.11	787.42	6.32	3.93	226.23	157.48	287.94	188.23	1862.04	262.85	0.53	25.01	7.00	15.00
	12.0	47.017	59.89	2319.87	885.70	6.22	3.85	257.76	177.14	333.01	216.31	2129.58	295.72	0.53	21.27	5.83	12.50
	12.5	48.728	62.07	2384.91	907.60	6.20	3.82	264.99	181.52	343.65	222.86	2191.20	303.07	0.53	20.52	5.50	11.90
	12.5	56.563	67.07	3135.96	1003.95	6.84	3.87	313.60	200.79	408.23	244.74	2541.18	341.48	0.57	18.99	5.50	13.50
100 × 200	4.0	18.203	23.19	1222.63	416.24	7.26	4.24	122.26	83.25	150.39	92.84	983.43	141.86	0.59	54.94	21.00	46.00
	4.5	20.389	25.97	1360.27	461.47	7.24	4.22	136.03	92.29	167.85	103.46	1094.82	157.20	0.59	49.05	18.22	40.44
	5.0	22.554	28.73	1494.64	505.27	7.21	4.19	149.46	101.05	185.02	113.86	1203.68	172.04	0.59	44.34	16.00	36.00
	6.0	26.826	34.17	1753.66	588.62	7.16	4.15	175.37	117.72	218.48	134.01	1413.87	200.28	0.58	37.28	12.67	29.33
	6.3	28.092	35.79	1828.86	612.55	7.15	4.14	182.89	122.51	228.30	139.89	1474.98	208.38	0.58	35.60	12.87	28.75
	8.0	35.131	44.75	2233.60	739.01	7.06	4.06	223.36	147.80	281.95	171.78	1804.46	251.23	0.58	28.46	9.50	22.00
	9.0	39.164	49.89	2454.98	806.38	7.01	4.02	245.50	161.28	311.97	189.42	1985.02	274.09	0.58	25.53	8.11	19.22
	10.0	43.118	54.93	2664.25	868.76	6.96	3.98	266.43	173.75	340.87	206.23	2155.89	295.30	0.57	23.19	7.00	17.00
	12.0	50.785	64.69	3047.39	979.20	6.86	3.89	304.74	195.84	395.31	237.43	2468.86	332.99	0.57	19.69	5.83	14.17
100 × 300	4.0	16.947	21.59	944.51	379.35	6.61	4.19	104.95	75.87	128.00	85.16	852.07	127.11	0.55	59.01	21.00	41.00
	5.0	20.984	26.73	1152.67	460.10	6.57	4.15	128.07	92.02	157.29	104.36	1042.42	153.99	0.55	47.65	16.00	32.00
	6.0	24.942	31.77	1350.07	535.53	6.52	4.11	150.01	107.11	185.51	122.73	1223.86	179.06	0.54	40.09	12.67	26.00
	6.3	26.114	33.27	1407.22	557.15	6.50	4.09	156.36	111.43	193.77	128.09	1276.57	186.25	0.54	38.29	12.87	25.57
	8.0	32.619	41.55	1713.39	671.13	6.42	4.02	190.38	134.23	238.80	157.06	1560.32	224.13	0.54	30.66	9.50	19.50
	10.0	39.978	50.93	2036.11	787.42	6.32	3.93	226.23	157.48	287.94	188.23	1862.04	262.85	0.53	25.01	7.00	15.00
	12.0	47.017	59.89	2319.87	885.70	6.22	3.85	257.76	177.14	333.01	216.31	2129.58	295.72	0.53	21.27	5.83	12.50
	12.5	48.728	62.07	2384.91	907.60	6.20	3.82	264.99	181.52	343.65	222.86	2191.20	303.07	0.53	20.52	5.50	11.90
	12.5	56.563	67.07	3135.96	1003.95	6.84	3.87	313.60	200.79	408.23	244.74	2541.18	341.48	0.57	18.99	5.50	13.50
120 × 200	4.0	19.459	24.79	1376.31	625.65	7.45	5.02	137.63	104.27	166.07	116.83	1342.72	172.54	0.63	51.39	26.00	46.00
	5.0	24.124	30.73	1684.81	762.38	7.40											

Hot Finished Rectangular Hollow Sections

BS EN 10210

Designation		Weight	Cross Sectional Area	Second Moment of Area	Second Moment of Area	Radius of Gyration	Radius of Gyration	Elastic Modulus	Elastic Modulus	Plastic Modulus	Plastic Modulus	Torsional inertia contant	Torsional modulus contant	Superficial Area per metre length	Nominal Length per tonne	Ratio for Local Buckling	Ratio for Local Buckling
Size	Thickness																
B × H	T	M	A	I _{yy}	I _{zz}	i _{yy}	i _{zz}	W _{plly}	W _{elzz}	W _{plyy}	W _{plzz}	L _t	C _t	As	L	Flange	Web
mm	mm	Kg/m	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³	cm ³	cm ³	cm ⁴	cm ³	m ² /m	m	B/T	H/T
200 × 300	4.5	34.519	43.97	5727.93	3080.49	11.41	8.37	381.86	308.05	453.19	343.80	6175.34	500.96	0.99	28.97	40.44	62.67
	5.0	38.254	48.73	6322.27	3396.31	11.39	8.35	421.48	339.63	501.18	380.02	6824.16	551.99	0.99	26.14	36.00	56.00
	6.0	45.666	58.17	7486.26	4012.54	11.34	8.31	499.08	401.25	595.75	451.28	8099.60	651.41	0.98	21.90	29.33	46.00
	6.3	47.874	60.99	7829.08	4193.44	11.33	8.29	521.94	419.34	623.76	472.36	8476.47	680.56	0.98	20.89	28.75	44.62
	8.0	60.251	76.75	9716.68	5184.43	11.25	8.22	647.78	518.44	779.32	589.15	10562.12	839.96	0.98	16.60	22.00	34.50
	9.0	67.424	85.89	10783.84	5740.70	11.21	8.18	718.92	574.07	868.33	655.77	11749.45	929.25	0.98	14.83	19.22	30.33
	10.0	74.518	94.93	11819.43	6277.58	11.16	8.13	787.96	627.76	955.50	720.87	12907.66	1015.31	0.97	13.42	17.00	27.00
	12.0	88.465	112.69	13797.37	7294.43	11.06	8.05	919.82	729.44	1124.38	846.51	15137.15	1178.05	0.97	11.30	14.17	22.50
	12.5	91.903	117.07	14272.69	7537.01	11.04	8.02	951.51	753.70	1165.47	876.98	15676.50	1216.83	0.97	10.88	13.50	21.50
	16.0	115.405	147.01	17390.25	9108.84	10.88	7.87	1159.35	910.88	1440.54	1079.87	19251.64	1467.93	0.96	8.67	10.00	16.25
200 × 400	4.5	41.584	52.97	11434.11	3940.60	14.69	8.62	571.71	394.06	695.55	431.77	9176.30	672.94	1.19	24.05	40.44	84.89
	5.0	46.104	58.73	12635.66	4347.14	14.67	8.60	631.78	434.71	769.83	477.52	10144.00	742.11	1.19	21.69	36.00	76.00
	6.0	55.086	70.17	14998.11	5141.98	14.62	8.56	749.91	514.20	916.62	567.68	12048.54	877.22	1.18	18.15	29.33	62.67
	6.3	57.765	73.59	15696.33	5375.73	14.60	8.55	784.82	537.57	960.19	594.39	12611.90	916.92	1.18	17.31	28.75	60.49
	8.0	72.811	92.75	19562.02	6659.84	14.52	8.47	978.10	665.98	1203.08	742.75	15734.90	1134.86	1.18	13.73	22.00	47.00
	9.0	81.554	103.89	21764.38	7383.56	14.47	8.43	1088.22	738.36	1342.78	827.67	17517.10	1257.57	1.18	12.26	19.22	41.44
	10.0	90.218	114.93	23914.29	8084.25	14.43	8.39	1195.71	808.43	1480.14	910.87	19258.89	1376.31	1.17	11.08	17.00	37.00
	12.0	107.305	136.69	28058.57	9417.95	14.33	8.30	1402.93	941.80	1747.86	1072.11	22621.99	1602.22	1.17	9.32	14.17	30.83
	12.5	111.528	142.07	29062.56	9737.53	14.30	8.28	1453.13	973.75	1813.34	1111.35	23437.79	1656.34	1.17	8.97	13.50	29.50
	16.0	140.525	179.01	35737.62	11824.15	14.13	8.13	1786.88	1182.42	2255.60	1374.27	28871.29	2009.86	1.16	7.12	10.00	22.50
250 × 350	6.0	55.086	70.17	12616.32	7538.26	13.41	10.36	720.93	603.06	851.88	677.42	14528.89	967.01	1.18	18.15	37.67	54.33
	6.3	57.765	73.59	13202.92	7885.11	13.39	10.35	754.45	630.81	892.37	709.46	15214.51	1011.18	1.18	17.31	36.68	52.56
	8.0	72.811	92.75	16449.35	9798.26	13.32	10.28	939.96	783.86	1118.00	887.83	19026.91	1254.38	1.18	13.73	28.25	40.75
	10.0	90.218	114.93	20101.90	11936.88	13.23	10.19	1148.68	954.95	1375.32	1090.68	23354.44	1525.38	1.17	11.08	22.00	32.00
	12.0	107.305	136.69	23577.38	13957.08	13.13	10.10	1347.28	1116.57	1623.92	1286.05	27512.62	1780.64	1.17	9.32	18.33	26.67
	12.5	111.528	142.07	24418.99	14443.97	13.11	10.08	1395.37	1155.52	1684.72	1333.72	28525.82	1842.06	1.17	8.97	17.50	25.50
	16.0	140.525	179.01	30010.76	17654.02	12.95	9.93	1714.90	1412.32	2095.27	1654.60	35325.18	2246.24	1.16	7.12	13.13	19.38
300 × 400	6.3	67.656	86.19	20579.24	13264.27	15.45	12.41	1028.96	884.28	1208.22	993.82	24739.51	1404.81	1.38	14.78	44.62	60.49
	8.0	85.371	108.75	25709.43	16539.51	15.38	12.33	1285.47	1102.63	1516.68	1246.52	31014.00	1748.80	1.38	11.71	34.50	47.00
	9.0	95.684	121.89	28645.24	18407.56	15.33	12.29	1432.26	1227.17	1694.68	1392.13	34625.93	1944.52	1.38	10.45	30.33	41.44
	10.0	105.918	134.93	31520.95	20232.76	15.28	12.25	1576.05	1348.85	1870.14	1535.50	38179.57	2135.45	1.37	9.44	27.00	37.00
	12.0	126.145	160.69	37094.09	23756.41	15.19	12.16	1854.70	1583.76	2213.46	1815.58	45112.66	2503.20	1.37	7.93	22.50	30.83
	12.5	131.153	167.07	38450.58	24611.23	15.17	12.14	1922.53	1640.75	2297.71	1884.22	46809.76	2592.25	1.37	7.62	21.50	29.50
	16.0	165.645	211.01	47540.93	30308.87	15.01	11.98	2377.05	2020.59	2870.00	2349.34	58286.43	3184.42	1.36	6.04	16.25	22.50